2. Colección canónica LR(1)

Reglas de construcción

Un ítem es $[A \to \alpha \cdot \beta, a]$. En los datos se representa como `[produccion, posicion, lookahead]`; la posición indica cuántos símbolos del RHS están antes del punto.

$$I_0 = \text{CLOSURE}(\{[\texttt{Start} \to \cdot\texttt{Program}, \$]\}).$$

Para $[A \to \alpha \cdot B\beta, a]$, CLOSURE agrega $[B \to \cdot\gamma, b]$ por cada producción $B \to \gamma$ y cada $b \in \text{FIRST}(\beta a)$. Repetir hasta no agregar ítems. GOTO mueve el punto sobre el símbolo y cierra el resultado. Solo se crea un estado nuevo si el conjunto completo de ítems LR(1) es distinto.

```
I0 = { [Start -> . Program, $],
       [Program -> . int main ( ) Block, $] }
GOTO(I0, int) = I2
GOTO(I0, Program) = I1
I1 = { [Start -> Program ., $] }  // aceptacion
```

En I28 aparece `Decl -> Type ID . Init ;`. CLOSURE usa $\text{FIRST}(;a) = \{;\}$ y agrega `Init -> . = Expr` e `Init -> .`, ambos con lookahead `;`.

Elección de CLR(1)

Se mantiene el método acordado: construir la colección canónica, calcular $\text{FIRST}(\beta a)$ y conservar los lookaheads de cada estado. No se fusionan estados por núcleo LR(0). La colección mantiene los lookaheads propios de cada estado, conforme al método CLR(1) acordado para el proyecto.

Resultado

161 estados alcanzables y distintos; 730 transiciones. El ítem de aceptación está en I1. Un ítem con posición igual a la longitud del RHS está completo; p0 es aceptación y las demás producciones generan reducciones bajo su lookahead.

Ejemplo de CLOSURE y GOTO en I28

Se parte del ítem `[Decl -> Type ID . Init ;, a]`. El no terminal después del punto es Init; la secuencia que queda después de él es `;`. Por tanto:

$$B = \texttt{Init}, \qquad \beta = \texttt{;}, \qquad \text{FIRST}(\beta a) = \text{FIRST}(\texttt{;}a) = \{\texttt{;}\}.$$

1. Consultar las dos producciones de Init: p11 y p12.
2. Agregar `[Init -> . = Expr, ;]` y `[Init -> ., ;]`.
3. El primer ítem permite avanzar con `=` hasta I49. El segundo ya está completo porque p12 tiene RHS vacío: permite reducir únicamente ante `;`.
4. Al reconocer Init, GOTO(I28,Init) lleva a I50 y el ítem de Decl queda como `Decl -> Type ID Init . ;`.

La anticipación `;` de los ítems añadidos proviene de FIRST(βa). No se obtiene copiando sin más `a`, ni calculando FOLLOW(Init).

Qué pasa si la secuencia puede ser vacía

Al entrar al bloque, I7 contiene `[Block -> { . Items }, $]`. Al expandir Items se usa FIRST de `}$`, cuyo único inicio es `}`. Por eso aparecen inicialmente ítems de Items con ese lookahead.

Después, `[Items -> . Items Item, }]` vuelve a expandir Items. Esta vez se calcula FIRST(`Item}`): se añaden los posibles inicios de Item. La repetición de CLOSURE reúne todos esos contextos hasta llegar al punto fijo. Si una secuencia β es anulable, el cálculo debe continuar hasta `a`; si β está vacía, FIRST(βa) es {a}.

Por qué no se eliminan estados de igual núcleo

I9 e I122 contienen el mismo núcleo `Simple -> ; .`. Sin embargo, I122 incluye el lookahead `else` e I9 no. En I122 se puede reducir p24 al ver else; en I9 esa reducción no está autorizada. Son estados CLR(1) distintos, aunque su parte LR(0) coincida.

Cómo leer la colección compacta

`(12,0,L3)*` significa p12, punto en posición cero y todos los lookaheads de L3. Como p12 es vacía y L3 contiene solo `;`, equivale a `[Init -> ., ;]`. El asterisco marca que está completo. En cambio, `(7,2,L1)` tiene dos símbolos reconocidos de `Decl -> Type ID Init ;`: el punto está después de ID.

Los conjuntos L0–L13 solo evitan repetir listas de lookaheads en el documento. Cada estado sigue representando su conjunto completo de ítems LR(1), sin fusión de estados.

FIRST calculados

No terminal(es)	FIRST
Add, Equality, Expr, Mul, Relation, Unary	!, (, +, -, CHAR, FLOAT, ID, INT, STRING
Block	{
Decl, Type	char, float, int
Init	epsilon, =
Item	;, ID, char, float, if, int, return, while, {
Items	;, epsilon, ID, char, float, if, int, return, while, {
Matched, Stmt	;, ID, if, return, while, {
Primary	(, CHAR, FLOAT, ID, INT, STRING
Program, Start	int
Simple	;, ID, return
Unmatched	if, while

Colección completa en notación compacta

La terna `(p,d,Lk)` representa los ítems `[p,d,a]` para todos los lookaheads `a` del conjunto `Lk`. Por ejemplo, `(0,0,L0)` usa p0 y el punto al inicio. Agrupar lookaheads dentro de un mismo estado es solo notación: no fusiona estados CLR(1). Los IDs de producción son los del documento 1. El asterisco identifica un ítem completo; en I1 corresponde a aceptación.

Alias	Lookaheads
L0	$
L1	;, ID, char, float, if, int, return, while, {, }
L2	ID
L3	;
L4	!=, ;, ==
L5	!=, ;, <, <=, ==, >, >=
L6	!=, +, -, ;, <, <=, ==, >, >=
L7	!=, %, *, +, -, /, ;, <, <=, ==, >, >=
L8	)
L9	!=,), ==
L10	!=,), <, <=, ==, >, >=
L11	!=,), +, -, <, <=, ==, >, >=
L12	!=, %,), *, +, -, /, <, <=, ==, >, >=
L13	;, ID, char, else, float, if, int, return, while, {, }

Estado	Ítems: (producción, posición, conjunto de lookaheads)
I0	(0,0,L0); (1,0,L0)
I1	(0,1,L0)*
I2	(1,1,L0)
I3	(1,2,L0)
I4	(1,3,L0)
I5	(1,4,L0); (2,0,L0)

Estado	Ítems: (producción, posición, conjunto de lookaheads)
I6	(1,5,L0)*
I7	(2,1,L0); (3,0,L1); (4,0,L1)*
I8	(2,0,L1); (2,2,L0); (3,1,L1); (5,0,L1); (6,0,L1); (7,0,L1); (8,0,L2); (9,0,L2); (10,0,L2); (13,0,L1); (14,0,L1); (15,0,L1); (16,0,L1); (17,0,L1); (18,0,L1); (19,0,L1); (20,0,L1); (21,0,L1); (22,0,L1); (23,0,L1); (24,0,L1)
I9	(24,1,L1)*
I10	(16,1,L1)*
I11	(5,1,L1)*
I12	(22,1,L1)
I13	(3,2,L1)*
I14	(13,1,L1)*
I15	(15,1,L1)*
I16	(6,1,L1)*
I17	(7,1,L1)
I18	(14,1,L1)*
I19	(10,1,L2)*
I20	(9,1,L2)*
I21	(17,1,L1); (19,1,L1); (20,1,L1)
I22	(8,1,L2)*
I23	(23,1,L1); (25,0,L3); (26,0,L4); (27,0,L4); (28,0,L4); (29,0,L5); (30,0,L5); (31,0,L5); (32,0,L5); (33,0,L5); (34,0,L6); (35,0,L6); (36,0,L6); (37,0,L7); (38,0,L7); (39,0,L7); (40,0,L7); (41,0,L7); (42,0,L7); (43,0,L7); (44,0,L7); (45,0,L7); (46,0,L7); (47,0,L7); (48,0,L7); (49,0,L7); (50,0,L7)
I24	(18,1,L1); (21,1,L1)
I25	(2,1,L1); (3,0,L1); (4,0,L1)*
I26	(2,3,L0)*
I27	(22,2,L1); (25,0,L3); (26,0,L4); (27,0,L4); (28,0,L4); (29,0,L5); (30,0,L5); (31,0,L5); (32,0,L5); (33,0,L5); (34,0,L6); (35,0,L6); (36,0,L6); (37,0,L7); (38,0,L7); (39,0,L7); (40,0,L7); (41,0,L7); (42,0,L7); (43,0,L7); (44,0,L7); (45,0,L7); (46,0,L7); (47,0,L7); (48,0,L7); (49,0,L7); (50,0,L7)
I28	(7,2,L1); (11,0,L3); (12,0,L3)*
I29	(17,2,L1); (19,2,L1); (20,2,L1); (25,0,L8); (26,0,L9); (27,0,L9); (28,0,L9); (29,0,L10); (30,0,L10); (31,0,L10); (32,0,L10); (33,0,L10); (34,0,L11); (35,0,L11); (36,0,L11); (37,0,L12); (38,0,L12); (39,0,L12); (40,0,L12); (41,0,L12); (42,0,L12); (43,0,L12); (44,0,L12); (45,0,L12); (46,0,L12); (47,0,L12); (48,0,L12); (49,0,L12); (50,0,L12)
I30	(41,0,L7); (42,0,L7); (43,0,L7); (43,1,L7); (44,0,L7); (45,0,L7); (46,0,L7); (47,0,L7); (48,0,L7); (49,0,L7); (50,0,L7)
I31	(25,0,L8); (26,0,L9); (27,0,L9); (28,0,L9); (29,0,L10); (30,0,L10); (31,0,L10); (32,0,L10); (33,0,L10); (34,0,L11); (35,0,L11); (36,0,L11); (37,0,L12); (38,0,L12); (39,0,L12); (40,0,L12); (41,0,L12); (42,0,L12); (43,0,L12); (44,0,L12); (45,0,L12); (46,0,L12); (47,0,L12); (48,0,L12); (49,0,L12); (50,0,L12); (50,1,L7)
I32	(41,0,L7); (41,1,L7); (42,0,L7); (43,0,L7); (44,0,L7); (45,0,L7); (46,0,L7); (47,0,L7); (48,0,L7); (49,0,L7); (50,0,L7)

Estado	Ítems: (producción, posición, conjunto de lookaheads)
I33	(41,0,L7); (42,0,L7); (42,1,L7); (43,0,L7); (44,0,L7); (45,0,L7); (46,0,L7); (47,0,L7); (48,0,L7); (49,0,L7); (50,0,L7)
I34	(33,1,L5)*; (34,1,L6); (35,1,L6)
I35	(48,1,L7)*
I36	(25,1,L3)*; (26,1,L4); (27,1,L4)
I37	(23,2,L1)
I38	(47,1,L7)*
I39	(45,1,L7)*
I40	(46,1,L7)*
I41	(36,1,L6)*; (37,1,L7); (38,1,L7); (39,1,L7)
I42	(44,1,L7)*
I43	(28,1,L4)*; (29,1,L5); (30,1,L5); (31,1,L5); (32,1,L5)
I44	(49,1,L7)*
I45	(40,1,L7)*
I46	(18,2,L1); (21,2,L1); (25,0,L8); (26,0,L9); (27,0,L9); (28,0,L9); (29,0,L10); (30,0,L10); (31,0,L10); (32,0,L10); (33,0,L10); (34,0,L11); (35,0,L11); (36,0,L11); (37,0,L12); (38,0,L12); (39,0,L12); (40,0,L12); (41,0,L12); (42,0,L12); (43,0,L12); (44,0,L12); (45,0,L12); (46,0,L12); (47,0,L12); (48,0,L12); (49,0,L12); (50,0,L12)
I47	(2,0,L1); (2,2,L1); (3,1,L1); (5,0,L1); (6,0,L1); (7,0,L1); (8,0,L2); (9,0,L2); (10,0,L2); (13,0,L1); (14,0,L1); (15,0,L1); (16,0,L1); (17,0,L1); (18,0,L1); (19,0,L1); (20,0,L1); (21,0,L1); (22,0,L1); (23,0,L1); (24,0,L1)
I48	(22,3,L1)
I49	(11,1,L3); (25,0,L3); (26,0,L4); (27,0,L4); (28,0,L4); (29,0,L5); (30,0,L5); (31,0,L5); (32,0,L5); (33,0,L5); (34,0,L6); (35,0,L6); (36,0,L6); (37,0,L7); (38,0,L7); (39,0,L7); (40,0,L7); (41,0,L7); (42,0,L7); (43,0,L7); (44,0,L7); (45,0,L7); (46,0,L7); (47,0,L7); (48,0,L7); (49,0,L7); (50,0,L7)
I50	(7,3,L1)
I51	(41,0,L12); (42,0,L12); (43,0,L12); (43,1,L12); (44,0,L12); (45,0,L12); (46,0,L12); (47,0,L12); (48,0,L12); (49,0,L12); (50,0,L12)
I52	(25,0,L8); (26,0,L9); (27,0,L9); (28,0,L9); (29,0,L10); (30,0,L10); (31,0,L10); (32,0,L10); (33,0,L10); (34,0,L11); (35,0,L11); (36,0,L11); (37,0,L12); (38,0,L12); (39,0,L12); (40,0,L12); (41,0,L12); (42,0,L12); (43,0,L12); (44,0,L12); (45,0,L12); (46,0,L12); (47,0,L12); (48,0,L12); (49,0,L12); (50,0,L12); (50,1,L12)
I53	(41,0,L12); (41,1,L12); (42,0,L12); (43,0,L12); (44,0,L12); (45,0,L12); (46,0,L12); (47,0,L12); (48,0,L12); (49,0,L12); (50,0,L12)
I54	(41,0,L12); (42,0,L12); (42,1,L12); (43,0,L12); (44,0,L12); (45,0,L12); (46,0,L12); (47,0,L12); (48,0,L12); (49,0,L12); (50,0,L12)
I55	(33,1,L10)*; (34,1,L11); (35,1,L11)
I56	(48,1,L12)*
I57	(25,1,L8)*; (26,1,L9); (27,1,L9)
I58	(17,3,L1); (19,3,L1); (20,3,L1)
I59	(47,1,L12)*
I60	(45,1,L12)*
I61	(46,1,L12)*

Estado	Ítems: (producción, posición, conjunto de lookaheads)
I62	(36,1,L11)*; (37,1,L12); (38,1,L12); (39,1,L12)
I63	(44,1,L12)*
I64	(28,1,L9)*; (29,1,L10); (30,1,L10); (31,1,L10); (32,1,L10)
I65	(49,1,L12)*
I66	(40,1,L12)*
I67	(43,2,L7)*
I68	(50,2,L7)
I69	(41,2,L7)*
I70	(42,2,L7)*
I71	(34,2,L6); (37,0,L7); (38,0,L7); (39,0,L7); (40,0,L7); (41,0,L7); (42,0,L7); (43,0,L7); (44,0,L7); (45,0,L7); (46,0,L7); (47,0,L7); (48,0,L7); (49,0,L7); (50,0,L7)
I72	(35,2,L6); (37,0,L7); (38,0,L7); (39,0,L7); (40,0,L7); (41,0,L7); (42,0,L7); (43,0,L7); (44,0,L7); (45,0,L7); (46,0,L7); (47,0,L7); (48,0,L7); (49,0,L7); (50,0,L7)
I73	(27,2,L4); (29,0,L5); (30,0,L5); (31,0,L5); (32,0,L5); (33,0,L5); (34,0,L6); (35,0,L6); (36,0,L6); (37,0,L7); (38,0,L7); (39,0,L7); (40,0,L7); (41,0,L7); (42,0,L7); (43,0,L7); (44,0,L7); (45,0,L7); (46,0,L7); (47,0,L7); (48,0,L7); (49,0,L7); (50,0,L7)
I74	(26,2,L4); (29,0,L5); (30,0,L5); (31,0,L5); (32,0,L5); (33,0,L5); (34,0,L6); (35,0,L6); (36,0,L6); (37,0,L7); (38,0,L7); (39,0,L7); (40,0,L7); (41,0,L7); (42,0,L7); (43,0,L7); (44,0,L7); (45,0,L7); (46,0,L7); (47,0,L7); (48,0,L7); (49,0,L7); (50,0,L7)
I75	(23,3,L1)*
I76	(39,2,L7); (41,0,L7); (42,0,L7); (43,0,L7); (44,0,L7); (45,0,L7); (46,0,L7); (47,0,L7); (48,0,L7); (49,0,L7); (50,0,L7)
I77	(37,2,L7); (41,0,L7); (42,0,L7); (43,0,L7); (44,0,L7); (45,0,L7); (46,0,L7); (47,0,L7); (48,0,L7); (49,0,L7); (50,0,L7)
I78	(38,2,L7); (41,0,L7); (42,0,L7); (43,0,L7); (44,0,L7); (45,0,L7); (46,0,L7); (47,0,L7); (48,0,L7); (49,0,L7); (50,0,L7)
I79	(29,2,L5); (34,0,L6); (35,0,L6); (36,0,L6); (37,0,L7); (38,0,L7); (39,0,L7); (40,0,L7); (41,0,L7); (42,0,L7); (43,0,L7); (44,0,L7); (45,0,L7); (46,0,L7); (47,0,L7); (48,0,L7); (49,0,L7); (50,0,L7)
I80	(30,2,L5); (34,0,L6); (35,0,L6); (36,0,L6); (37,0,L7); (38,0,L7); (39,0,L7); (40,0,L7); (41,0,L7); (42,0,L7); (43,0,L7); (44,0,L7); (45,0,L7); (46,0,L7); (47,0,L7); (48,0,L7); (49,0,L7); (50,0,L7)
I81	(31,2,L5); (34,0,L6); (35,0,L6); (36,0,L6); (37,0,L7); (38,0,L7); (39,0,L7); (40,0,L7); (41,0,L7); (42,0,L7); (43,0,L7); (44,0,L7); (45,0,L7); (46,0,L7); (47,0,L7); (48,0,L7); (49,0,L7); (50,0,L7)
I82	(32,2,L5); (34,0,L6); (35,0,L6); (36,0,L6); (37,0,L7); (38,0,L7); (39,0,L7); (40,0,L7); (41,0,L7); (42,0,L7); (43,0,L7); (44,0,L7); (45,0,L7); (46,0,L7); (47,0,L7); (48,0,L7); (49,0,L7); (50,0,L7)
I83	(18,3,L1); (21,3,L1)
I84	(2,3,L1)*
I85	(22,4,L1)*
I86	(11,2,L3)*
I87	(7,4,L1)*
I88	(43,2,L12)*
I89	(50,2,L12)

Estado	Ítems: (producción, posición, conjunto de lookaheads)
I90	(41,2,L12)*
I91	(42,2,L12)*
I92	(34,2,L11); (37,0,L12); (38,0,L12); (39,0,L12); (40,0,L12); (41,0,L12); (42,0,L12); (43,0,L12); (44,0,L12); (45,0,L12); (46,0,L12); (47,0,L12); (48,0,L12); (49,0,L12); (50,0,L12)
I93	(35,2,L11); (37,0,L12); (38,0,L12); (39,0,L12); (40,0,L12); (41,0,L12); (42,0,L12); (43,0,L12); (44,0,L12); (45,0,L12); (46,0,L12); (47,0,L12); (48,0,L12); (49,0,L12); (50,0,L12)
I94	(27,2,L9); (29,0,L10); (30,0,L10); (31,0,L10); (32,0,L10); (33,0,L10); (34,0,L11); (35,0,L11); (36,0,L11); (37,0,L12); (38,0,L12); (39,0,L12); (40,0,L12); (41,0,L12); (42,0,L12); (43,0,L12); (44,0,L12); (45,0,L12); (46,0,L12); (47,0,L12); (48,0,L12); (49,0,L12); (50,0,L12)
I95	(26,2,L9); (29,0,L10); (30,0,L10); (31,0,L10); (32,0,L10); (33,0,L10); (34,0,L11); (35,0,L11); (36,0,L11); (37,0,L12); (38,0,L12); (39,0,L12); (40,0,L12); (41,0,L12); (42,0,L12); (43,0,L12); (44,0,L12); (45,0,L12); (46,0,L12); (47,0,L12); (48,0,L12); (49,0,L12); (50,0,L12)
I96	(2,0,L13); (13,0,L1); (14,0,L1); (15,0,L13); (16,0,L13); (17,0,L13); (17,4,L1); (18,0,L13); (19,0,L1); (19,4,L1); (20,0,L1); (20,4,L1); (21,0,L1); (22,0,L13); (23,0,L13); (24,0,L13)
I97	(39,2,L12); (41,0,L12); (42,0,L12); (43,0,L12); (44,0,L12); (45,0,L12); (46,0,L12); (47,0,L12); (48,0,L12); (49,0,L12); (50,0,L12)
I98	(37,2,L12); (41,0,L12); (42,0,L12); (43,0,L12); (44,0,L12); (45,0,L12); (46,0,L12); (47,0,L12); (48,0,L12); (49,0,L12); (50,0,L12)
I99	(38,2,L12); (41,0,L12); (42,0,L12); (43,0,L12); (44,0,L12); (45,0,L12); (46,0,L12); (47,0,L12); (48,0,L12); (49,0,L12); (50,0,L12)
I100	(29,2,L10); (34,0,L11); (35,0,L11); (36,0,L11); (37,0,L12); (38,0,L12); (39,0,L12); (40,0,L12); (41,0,L12); (42,0,L12); (43,0,L12); (44,0,L12); (45,0,L12); (46,0,L12); (47,0,L12); (48,0,L12); (49,0,L12); (50,0,L12)
I101	(30,2,L10); (34,0,L11); (35,0,L11); (36,0,L11); (37,0,L12); (38,0,L12); (39,0,L12); (40,0,L12); (41,0,L12); (42,0,L12); (43,0,L12); (44,0,L12); (45,0,L12); (46,0,L12); (47,0,L12); (48,0,L12); (49,0,L12); (50,0,L12)
I102	(31,2,L10); (34,0,L11); (35,0,L11); (36,0,L11); (37,0,L12); (38,0,L12); (39,0,L12); (40,0,L12); (41,0,L12); (42,0,L12); (43,0,L12); (44,0,L12); (45,0,L12); (46,0,L12); (47,0,L12); (48,0,L12); (49,0,L12); (50,0,L12)
I103	(32,2,L10); (34,0,L11); (35,0,L11); (36,0,L11); (37,0,L12); (38,0,L12); (39,0,L12); (40,0,L12); (41,0,L12); (42,0,L12); (43,0,L12); (44,0,L12); (45,0,L12); (46,0,L12); (47,0,L12); (48,0,L12); (49,0,L12); (50,0,L12)
I104	(50,3,L7)*
I105	(34,3,L6)*; (37,1,L7); (38,1,L7); (39,1,L7)
I106	(35,3,L6)*; (37,1,L7); (38,1,L7); (39,1,L7)
I107	(27,3,L4)*; (29,1,L5); (30,1,L5); (31,1,L5); (32,1,L5)
I108	(26,3,L4)*; (29,1,L5); (30,1,L5); (31,1,L5); (32,1,L5)
I109	(39,3,L7)*
I110	(37,3,L7)*
I111	(38,3,L7)*
I112	(29,3,L5)*; (34,1,L6); (35,1,L6)

Estado	Ítems: (producción, posición, conjunto de lookaheads)
I113	(30,3,L5)*; (34,1,L6); (35,1,L6)
I114	(31,3,L5)*; (34,1,L6); (35,1,L6)
I115	(32,3,L5)*; (34,1,L6); (35,1,L6)
I116	(2,0,L1); (15,0,L1); (16,0,L1); (17,0,L1); (18,0,L1); (18,4,L1); (19,0,L1); (20,0,L1); (21,0,L1); (21,4,L1); (22,0,L1); (23,0,L1); (24,0,L1)
I117	(50,3,L12)*
I118	(34,3,L11)*; (37,1,L12); (38,1,L12); (39,1,L12)
I119	(35,3,L11)*; (37,1,L12); (38,1,L12); (39,1,L12)
I120	(27,3,L9)*; (29,1,L10); (30,1,L10); (31,1,L10); (32,1,L10)
I121	(26,3,L9)*; (29,1,L10); (30,1,L10); (31,1,L10); (32,1,L10)
I122	(24,1,L13)*
I123	(16,1,L13)*
I124	(22,1,L13)
I125	(13,1,L1)*; (17,5,L1); (20,5,L1)
I126	(15,1,L13)*
I127	(19,5,L1)*
I128	(17,1,L13); (19,1,L1); (20,1,L1)
I129	(23,1,L13); (25,0,L3); (26,0,L4); (27,0,L4); (28,0,L4); (29,0,L5); (30,0,L5); (31,0,L5); (32,0,L5); (33,0,L5); (34,0,L6); (35,0,L6); (36,0,L6); (37,0,L7); (38,0,L7); (39,0,L7); (40,0,L7); (41,0,L7); (42,0,L7); (43,0,L7); (44,0,L7); (45,0,L7); (46,0,L7); (47,0,L7); (48,0,L7); (49,0,L7); (50,0,L7)
I130	(18,1,L13); (21,1,L1)
I131	(2,1,L13); (3,0,L1); (4,0,L1)*
I132	(39,3,L12)*
I133	(37,3,L12)*
I134	(38,3,L12)*
I135	(29,3,L10)*; (34,1,L11); (35,1,L11)
I136	(30,3,L10)*; (34,1,L11); (35,1,L11)
I137	(31,3,L10)*; (34,1,L11); (35,1,L11)
I138	(32,3,L10)*; (34,1,L11); (35,1,L11)
I139	(18,5,L1)*
I140	(21,5,L1)*
I141	(22,2,L13); (25,0,L3); (26,0,L4); (27,0,L4); (28,0,L4); (29,0,L5); (30,0,L5); (31,0,L5); (32,0,L5); (33,0,L5); (34,0,L6); (35,0,L6); (36,0,L6); (37,0,L7); (38,0,L7); (39,0,L7); (40,0,L7); (41,0,L7); (42,0,L7); (43,0,L7); (44,0,L7); (45,0,L7); (46,0,L7); (47,0,L7); (48,0,L7); (49,0,L7); (50,0,L7)
I142	(2,0,L1); (15,0,L1); (16,0,L1); (17,0,L1); (17,6,L1); (18,0,L1); (19,0,L1); (20,0,L1); (20,6,L1); (21,0,L1); (22,0,L1); (23,0,L1); (24,0,L1)
I143	(17,2,L13); (19,2,L1); (20,2,L1); (25,0,L8); (26,0,L9); (27,0,L9); (28,0,L9); (29,0,L10); (30,0,L10); (31,0,L10); (32,0,L10); (33,0,L10); (34,0,L11); (35,0,L11); (36,0,L11); (37,0,L12); (38,0,L12); (39,0,L12); (40,0,L12); (41,0,L12); (42,0,L12); (43,0,L12); (44,0,L12); (45,0,L12); (46,0,L12); (47,0,L12); (48,0,L12); (49,0,L12); (50,0,L12)
I144	(23,2,L13)

Estado	Ítems: (producción, posición, conjunto de lookaheads)
I145	(18,2,L13); (21,2,L1); (25,0,L8); (26,0,L9); (27,0,L9); (28,0,L9); (29,0,L10); (30,0,L10); (31,0,L10); (32,0,L10); (33,0,L10); (34,0,L11); (35,0,L11); (36,0,L11); (37,0,L12); (38,0,L12); (39,0,L12); (40,0,L12); (41,0,L12); (42,0,L12); (43,0,L12); (44,0,L12); (45,0,L12); (46,0,L12); (47,0,L12); (48,0,L12); (49,0,L12); (50,0,L12)
I146	(2,0,L1); (2,2,L13); (3,1,L1); (5,0,L1); (6,0,L1); (7,0,L1); (8,0,L2); (9,0,L2); (10,0,L2); (13,0,L1); (14,0,L1); (15,0,L1); (16,0,L1); (17,0,L1); (18,0,L1); (19,0,L1); (20,0,L1); (21,0,L1); (22,0,L1); (23,0,L1); (24,0,L1)
I147	(22,3,L13)
I148	(17,7,L1)*
I149	(20,7,L1)*
I150	(17,3,L13); (19,3,L1); (20,3,L1)
I151	(23,3,L13)*
I152	(18,3,L13); (21,3,L1)
I153	(2,3,L13)*
I154	(22,4,L13)*
I155	(2,0,L13); (13,0,L1); (14,0,L1); (15,0,L13); (16,0,L13); (17,0,L13); (17,4,L13); (18,0,L13); (19,0,L1); (19,4,L1); (20,0,L1); (20,4,L1); (21,0,L1); (22,0,L13); (23,0,L13); (24,0,L13)
I156	(2,0,L13); (15,0,L13); (16,0,L13); (17,0,L13); (18,0,L13); (18,4,L13); (19,0,L1); (20,0,L1); (21,0,L1); (21,4,L1); (22,0,L13); (23,0,L13); (24,0,L13)
I157	(13,1,L1)*; (17,5,L13); (20,5,L1)
I158	(18,5,L13)*
I159	(2,0,L13); (15,0,L13); (16,0,L13); (17,0,L13); (17,6,L13); (18,0,L13); (19,0,L1); (20,0,L1); (20,6,L1); (21,0,L1); (22,0,L13); (23,0,L13); (24,0,L13)
I160	(17,7,L13)*

Todas las transiciones GOTO

Origen	Símbolo : destino
I0	`Program`: I1; `int`: I2
I1	Sin transiciones.
I2	`main`: I3
I3	`(`: I4
I4	`)`: I5
I5	`Block`: I6; `{`: I7
I6	Sin transiciones.
I7	`Items`: I8
I8	`;`: I9; `Block`: I10; `Decl`: I11; `ID`: I12; `Item`: I13; `Matched`: I14; `Simple`: I15; `Stmt`: I16; `Type`: I17; `Unmatched`: I18; `char`: I19; `float`: I20; `if`: I21; `int`: I22; `return`: I23; `while`: I24; `{`: I25; `}`: I26
I9	Sin transiciones.
I10	Sin transiciones.
I11	Sin transiciones.

Origen	Símbolo : destino
I12	=: I27
I13	Sin transiciones.
I14	Sin transiciones.
I15	Sin transiciones.
I16	Sin transiciones.
I17	ID: I28
I18	Sin transiciones.
I19	Sin transiciones.
I20	Sin transiciones.
I21	(: I29
I22	Sin transiciones.
I23	!: I30; (: I31; +: I32; -: I33; Add: I34; CHAR: I35; Equality: I36; Expr: I37; FLOAT: I38; ID: I39; INT: I40; Mul: I41; Primary: I42; Relation: I43; STRING: I44; Unary: I45
I24	(: I46
I25	Items: I47
I26	Sin transiciones.
I27	!: I30; (: I31; +: I32; -: I33; Add: I34; CHAR: I35; Equality: I36; Expr: I48; FLOAT: I38; ID: I39; INT: I40; Mul: I41; Primary: I42; Relation: I43; STRING: I44; Unary: I45
I28	=: I49; Init: I50
I29	!: I51; (: I52; +: I53; -: I54; Add: I55; CHAR: I56; Equality: I57; Expr: I58; FLOAT: I59; ID: I60; INT: I61; Mul: I62; Primary: I63; Relation: I64; STRING: I65; Unary: I66
I30	!: I30; (: I31; +: I32; -: I33; CHAR: I35; FLOAT: I38; ID: I39; INT: I40; Primary: I42; STRING: I44; Unary: I67
I31	!: I51; (: I52; +: I53; -: I54; Add: I55; CHAR: I56; Equality: I57; Expr: I68; FLOAT: I59; ID: I60; INT: I61; Mul: I62; Primary: I63; Relation: I64; STRING: I65; Unary: I66
I32	!: I30; (: I31; +: I32; -: I33; CHAR: I35; FLOAT: I38; ID: I39; INT: I40; Primary: I42; STRING: I44; Unary: I69
I33	!: I30; (: I31; +: I32; -: I33; CHAR: I35; FLOAT: I38; ID: I39; INT: I40; Primary: I42; STRING: I44; Unary: I70
I34	+: I71; -: I72
I35	Sin transiciones.
I36	!=: I73; ==: I74
I37	;: I75
I38	Sin transiciones.
I39	Sin transiciones.
I40	Sin transiciones.
I41	%: I76; *: I77; /: I78
I42	Sin transiciones.
I43	<: I79; <=: I80; >: I81; >=: I82
I44	Sin transiciones.
I45	Sin transiciones.
I46	!: I51; (: I52; +: I53; -: I54; Add: I55; CHAR: I56; Equality: I57; Expr: I83; FLOAT: I59; ID: I60; INT: I61; Mul: I62; Primary: I63; Relation: I64; STRING: I65; Unary: I66

Origen	Símbolo : destino
I47	`;`: I9; `Block`: I10; `Decl`: I11; `ID`: I12; `Item`: I13; `Matched`: I14; `Simple`: I15; `Stmt`: I16; `Type`: I17; `Unmatched`: I18; `char`: I19; `float`: I20; `if`: I21; `int`: I22; `return`: I23; `while`: I24; `{`: I25; `}`: I84
I48	`;`: I85
I49	`!`: I30; `(`: I31; `+`: I32; `-`: I33; `Add`: I34; `CHAR`: I35; `Equality`: I36; `Expr`: I86; `FLOAT`: I38; `ID`: I39; `INT`: I40; `Mul`: I41; `Primary`: I42; `Relation`: I43; `STRING`: I44; `Unary`: I45
I50	`;`: I87
I51	`!`: I51; `(`: I52; `+`: I53; `-`: I54; `CHAR`: I56; `FLOAT`: I59; `ID`: I60; `INT`: I61; `Primary`: I63; `STRING`: I65; `Unary`: I88
I52	`!`: I51; `(`: I52; `+`: I53; `-`: I54; `Add`: I55; `CHAR`: I56; `Equality`: I57; `Expr`: I89; `FLOAT`: I59; `ID`: I60; `INT`: I61; `Mul`: I62; `Primary`: I63; `Relation`: I64; `STRING`: I65; `Unary`: I66
I53	`!`: I51; `(`: I52; `+`: I53; `-`: I54; `CHAR`: I56; `FLOAT`: I59; `ID`: I60; `INT`: I61; `Primary`: I63; `STRING`: I65; `Unary`: I90
I54	`!`: I51; `(`: I52; `+`: I53; `-`: I54; `CHAR`: I56; `FLOAT`: I59; `ID`: I60; `INT`: I61; `Primary`: I63; `STRING`: I65; `Unary`: I91
I55	`+`: I92; `-`: I93
I56	Sin transiciones.
I57	`!=`: I94; `==`: I95
I58	`)`: I96
I59	Sin transiciones.
I60	Sin transiciones.
I61	Sin transiciones.
I62	`%`: I97; `*`: I98; `/`: I99
I63	Sin transiciones.
I64	`<`: I100; `<=`: I101; `>`: I102; `>=`: I103
I65	Sin transiciones.
I66	Sin transiciones.
I67	Sin transiciones.
I68	`)`: I104
I69	Sin transiciones.
I70	Sin transiciones.
I71	`!`: I30; `(`: I31; `+`: I32; `-`: I33; `CHAR`: I35; `FLOAT`: I38; `ID`: I39; `INT`: I40; `Mul`: I105; `Primary`: I42; `STRING`: I44; `Unary`: I45
I72	`!`: I30; `(`: I31; `+`: I32; `-`: I33; `CHAR`: I35; `FLOAT`: I38; `ID`: I39; `INT`: I40; `Mul`: I106; `Primary`: I42; `STRING`: I44; `Unary`: I45
I73	`!`: I30; `(`: I31; `+`: I32; `-`: I33; `Add`: I34; `CHAR`: I35; `FLOAT`: I38; `ID`: I39; `INT`: I40; `Mul`: I41; `Primary`: I42; `Relation`: I107; `STRING`: I44; `Unary`: I45
I74	`!`: I30; `(`: I31; `+`: I32; `-`: I33; `Add`: I34; `CHAR`: I35; `FLOAT`: I38; `ID`: I39; `INT`: I40; `Mul`: I41; `Primary`: I42; `Relation`: I108; `STRING`: I44; `Unary`: I45
I75	Sin transiciones.
I76	`!`: I30; `(`: I31; `+`: I32; `-`: I33; `CHAR`: I35; `FLOAT`: I38; `ID`: I39; `INT`: I40; `Primary`: I42; `STRING`: I44; `Unary`: I109
I77	`!`: I30; `(`: I31; `+`: I32; `-`: I33; `CHAR`: I35; `FLOAT`: I38; `ID`: I39; `INT`: I40; `Primary`: I42; `STRING`: I44; `Unary`: I110

Origen	Símbolo : destino
I78	`!`: I30; `(`: I31; `+`: I32; `-`: I33; `CHAR`: I35; `FLOAT`: I38; `ID`: I39; `INT`: I40; `Primary`: I42; `STRING`: I44; `Unary`: I111
I79	`!`: I30; `(`: I31; `+`: I32; `-`: I33; `Add`: I112; `CHAR`: I35; `FLOAT`: I38; `ID`: I39; `INT`: I40; `Mul`: I41; `Primary`: I42; `STRING`: I44; `Unary`: I45
I80	`!`: I30; `(`: I31; `+`: I32; `-`: I33; `Add`: I113; `CHAR`: I35; `FLOAT`: I38; `ID`: I39; `INT`: I40; `Mul`: I41; `Primary`: I42; `STRING`: I44; `Unary`: I45
I81	`!`: I30; `(`: I31; `+`: I32; `-`: I33; `Add`: I114; `CHAR`: I35; `FLOAT`: I38; `ID`: I39; `INT`: I40; `Mul`: I41; `Primary`: I42; `STRING`: I44; `Unary`: I45
I82	`!`: I30; `(`: I31; `+`: I32; `-`: I33; `Add`: I115; `CHAR`: I35; `FLOAT`: I38; `ID`: I39; `INT`: I40; `Mul`: I41; `Primary`: I42; `STRING`: I44; `Unary`: I45
I83	`)`: I116
I84	Sin transiciones.
I85	Sin transiciones.
I86	Sin transiciones.
I87	Sin transiciones.
I88	Sin transiciones.
I89	`)`: I117
I90	Sin transiciones.
I91	Sin transiciones.
I92	`!`: I51; `(`: I52; `+`: I53; `-`: I54; `CHAR`: I56; `FLOAT`: I59; `ID`: I60; `INT`: I61; `Mul`: I118; `Primary`: I63; `STRING`: I65; `Unary`: I66
I93	`!`: I51; `(`: I52; `+`: I53; `-`: I54; `CHAR`: I56; `FLOAT`: I59; `ID`: I60; `INT`: I61; `Mul`: I119; `Primary`: I63; `STRING`: I65; `Unary`: I66
I94	`!`: I51; `(`: I52; `+`: I53; `-`: I54; `Add`: I55; `CHAR`: I56; `FLOAT`: I59; `ID`: I60; `INT`: I61; `Mul`: I62; `Primary`: I63; `Relation`: I120; `STRING`: I65; `Unary`: I66
I95	`!`: I51; `(`: I52; `+`: I53; `-`: I54; `Add`: I55; `CHAR`: I56; `FLOAT`: I59; `ID`: I60; `INT`: I61; `Mul`: I62; `Primary`: I63; `Relation`: I121; `STRING`: I65; `Unary`: I66
I96	`;`: I122; `Block`: I123; `ID`: I124; `Matched`: I125; `Simple`: I126; `Stmt`: I127; `Unmatched`: I18; `if`: I128; `return`: I129; `while`: I130; `{`: I131
I97	`!`: I51; `(`: I52; `+`: I53; `-`: I54; `CHAR`: I56; `FLOAT`: I59; `ID`: I60; `INT`: I61; `Primary`: I63; `STRING`: I65; `Unary`: I132
I98	`!`: I51; `(`: I52; `+`: I53; `-`: I54; `CHAR`: I56; `FLOAT`: I59; `ID`: I60; `INT`: I61; `Primary`: I63; `STRING`: I65; `Unary`: I133
I99	`!`: I51; `(`: I52; `+`: I53; `-`: I54; `CHAR`: I56; `FLOAT`: I59; `ID`: I60; `INT`: I61; `Primary`: I63; `STRING`: I65; `Unary`: I134
I100	`!`: I51; `(`: I52; `+`: I53; `-`: I54; `Add`: I135; `CHAR`: I56; `FLOAT`: I59; `ID`: I60; `INT`: I61; `Mul`: I62; `Primary`: I63; `STRING`: I65; `Unary`: I66
I101	`!`: I51; `(`: I52; `+`: I53; `-`: I54; `Add`: I136; `CHAR`: I56; `FLOAT`: I59; `ID`: I60; `INT`: I61; `Mul`: I62; `Primary`: I63; `STRING`: I65; `Unary`: I66
I102	`!`: I51; `(`: I52; `+`: I53; `-`: I54; `Add`: I137; `CHAR`: I56; `FLOAT`: I59; `ID`: I60; `INT`: I61; `Mul`: I62; `Primary`: I63; `STRING`: I65; `Unary`: I66
I103	`!`: I51; `(`: I52; `+`: I53; `-`: I54; `Add`: I138; `CHAR`: I56; `FLOAT`: I59; `ID`: I60; `INT`: I61; `Mul`: I62; `Primary`: I63; `STRING`: I65; `Unary`: I66
I104	Sin transiciones.
I105	`%`: I76; `*`: I77; `/`: I78

Origen	Símbolo : destino
I106	%: I76; *: I77; /: I78
I107	<: I79; <=: I80; >: I81; >=: I82
I108	<: I79; <=: I80; >: I81; >=: I82
I109	Sin transiciones.
I110	Sin transiciones.
I111	Sin transiciones.
I112	+: I71; -: I72
I113	+: I71; -: I72
I114	+: I71; -: I72
I115	+: I71; -: I72
I116	;: I9; Block: I10; ID: I12; Matched: I139; Simple: I15; Unmatched: I140; if: I21; return: I23; while: I24; {: I25
I117	Sin transiciones.
I118	%: I97; *: I98; /: I99
I119	%: I97; *: I98; /: I99
I120	<: I100; <=: I101; >: I102; >=: I103
I121	<: I100; <=: I101; >: I102; >=: I103
I122	Sin transiciones.
I123	Sin transiciones.
I124	=: I141
I125	else: I142
I126	Sin transiciones.
I127	Sin transiciones.
I128	(: I143
I129	!: I30; (: I31; +: I32; -: I33; Add: I34; CHAR: I35; Equality: I36; Expr: I144; FLOAT: I38; ID: I39; INT: I40; Mul: I41; Primary: I42; Relation: I43; STRING: I44; Unary: I45
I130	(: I145
I131	Items: I146
I132	Sin transiciones.
I133	Sin transiciones.
I134	Sin transiciones.
I135	+: I92; -: I93
I136	+: I92; -: I93
I137	+: I92; -: I93
I138	+: I92; -: I93
I139	Sin transiciones.
I140	Sin transiciones.
I141	!: I30; (: I31; +: I32; -: I33; Add: I34; CHAR: I35; Equality: I36; Expr: I147; FLOAT: I38; ID: I39; INT: I40; Mul: I41; Primary: I42; Relation: I43; STRING: I44; Unary: I45
I142	;: I9; Block: I10; ID: I12; Matched: I148; Simple: I15; Unmatched: I149; if: I21; return: I23; while: I24; {: I25

Origen	Símbolo : destino
I143	`!`: I51; `(`: I52; `+`: I53; `-`: I54; `Add`: I55; `CHAR`: I56; `Equality`: I57; `Expr`: I150; `FLOAT`: I59; `ID`: I60; `INT`: I61; `Mul`: I62; `Primary`: I63; `Relation`: I64; `STRING`: I65; `Unary`: I66
I144	`;`: I151
I145	`!`: I51; `(`: I52; `+`: I53; `-`: I54; `Add`: I55; `CHAR`: I56; `Equality`: I57; `Expr`: I152; `FLOAT`: I59; `ID`: I60; `INT`: I61; `Mul`: I62; `Primary`: I63; `Relation`: I64; `STRING`: I65; `Unary`: I66
I146	`;`: I9; `Block`: I10; `Decl`: I11; `ID`: I12; `Item`: I13; `Matched`: I14; `Simple`: I15; `Stmt`: I16; `Type`: I17; `Unmatched`: I18; `char`: I19; `float`: I20; `if`: I21; `int`: I22; `return`: I23; `while`: I24; `{`: I25; `}`: I153
I147	`;`: I154
I148	Sin transiciones.
I149	Sin transiciones.
I150	`)`: I155
I151	Sin transiciones.
I152	`)`: I156
I153	Sin transiciones.
I154	Sin transiciones.
I155	`;`: I122; `Block`: I123; `ID`: I124; `Matched`: I157; `Simple`: I126; `Stmt`: I127; `Unmatched`: I18; `if`: I128; `return`: I129; `while`: I130; `{`: I131
I156	`;`: I122; `Block`: I123; `ID`: I124; `Matched`: I158; `Simple`: I126; `Unmatched`: I140; `if`: I128; `return`: I129; `while`: I130; `{`: I131
I157	`else`: I159
I158	Sin transiciones.
I159	`;`: I122; `Block`: I123; `ID`: I124; `Matched`: I160; `Simple`: I126; `Unmatched`: I149; `if`: I128; `return`: I129; `while`: I130; `{`: I131
I160	Sin transiciones.